

Measure of Central Tendency

① Mean

② Median

③ Mode

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Ages = {1, 3, 4, 5} → Distribution

$$\mu = \frac{1+3+4+5}{4} = \frac{13}{4} = 3.25$$

↓
 $\bar{x}$

② Median

Ages = {1, 3, 4, 5, 100} {Outlier}

Big Number
↓
Outlier

$$\mu = \frac{1+3+4+5+100}{5} = \frac{113}{5} = 22.6$$

↓
3.25

Ages = {4, 3, 1, 5, 100}

→ odd

↳ Sort the numbers {1, 3, 4, 5, 100, 200} → even

↳ Median → 4

$$\frac{4+5}{2} = 4.5$$

Good